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Quantum enabled cloud IOT collaboration for ultra low latency data processing in cyber physical systems

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Abstract
Cyber-Physical Systems (CPS) are evolving in a new era of intelligent infrastructures that combine sensing and computing. However, as quantum computing and ultra-dense IoT networks tends to get expanded, traditional encrypting data and optimizing systems are insufficient to secure the data in real time. Artificial Intelligence (AI) with post-quantum cryptography and edge intelligence transforms CPS into self-adaptive ecosystems that is capable of handling threats. However, the modern systems struggle with latency management and high computational complexity that occur after quantum computing. Hence, this paper research develops an AI-Driven Quantum-Resilient Cyber-Physical System (AI-QRCPS) framework that combines three novel layers and this includes: (i) a Post-Quantum Security Layer employing lattice-based cryptography and quantum key distribution for end-to-end confidentiality, (ii) an Adaptive Intelligence Layer using Reinforcement Learning (RL) and Firefly Optimization for dynamic resource orchestration and anomaly detection, and (iii) an IoT-Edge Coordination Layer for latency-aware task scheduling using double deep Q-learning. Experimental evaluation shows that the proposed framework achieves an average latency as low as 7.9–10.7 ms, throughput up to 94.5 tasks/s, energy consumption as low as 3.3–4.8 J/task, and task completion rates exceeding 98% across varying edge/IoT node densities and bandwidths.

Article Highlights

- The research designs a unified AI-driven quantum-resilient Cyber-Physical System (AI-QRCPS) framework.
- The results show an enhanced edge-IoT data transfer efficiency and intelligent load balancing, which gets maximized with relevance to the number of tasks processed per second under high-bandwidth conditions.
- The results show that the proposed framework is a flexible to build next-generation CPS networks.

Keywords Quantum-resilient CPS, Artificial Intelligence, Post-quantum cryptography, Cloud–IoT–Edge computing, Reinforcement learning

1 Introduction

1.1 Background

Cyber-Physical Systems (CPS) have emerged as the foundation of modern intelligent infrastructures, which enables tight Addition between computational algorithms, communication networks, and physical processes [1–15].

1.2 Literature survey

Goyal et al. [16] studies the impact of AI-enabled post-quantum CPS (QCPS) models. Shahab et al. [18] introduce a Quantum Machine Learning (QML) architecture that optimizes QoS while while reducing migration cost and energy consumption in CPS. Chandrasekar et al. [19] propose the Quantum-Resilient Hybrid Cryptographic Protocol (QHCP) that combines classical and post-quantum cryptography for long-term data integrity.

Several studies have explored nature-inspired and optimization-based approaches to improve CPS security and manage resources. Bahulayan et al. [17] introduce the Firefly Optimization Algorithm (FOA), which draws inspiration from swarm behavior, to boost security and save energy in CPS-IoT networks. Lu et al. [29] propose Non-collision Deterministic Scheduling (NDS) and Dynamic Queue Scheduling (DQS) techniques that specifically aimed at maintaining extremely low latency in time-critical networks.

The progression toward autonomous Industrial Cyber-Physical Systems (ICPS) is heavily supported by cloud-fog Addition and intelligent orchestration mechanisms. Jin et al. [20] advocate for Cloud-Fog Automation using the "3C co-design" approach, communication, computing, and control, to enhance autonomy, security, and QoS across industrial CPS. Fan et al. [21] extend this framework through a four-layer energy-efficient architecture is combined with a Double Deep Q-Learning (DDQ) algorithm for optimized communication.

In resource-intensive environments, Kamdjou et al. [22] provide an extensive analysis of performance optimization techniques for AR/VR-driven CPS utilizing Digital Twin (DT) models. Similarly, Elkhodr et al. [27] present the Combined Adaptive Security Framework for IoT (IASF-IoT), which uses AI, blockchain, quantum-resistant cryptography, and Digital Twins for predictive and proactive threat mitigation.

The emergence of 6G and Edge Computing has facilitated new directions for secure and real-time CPS communication. Alatawi [28] introduces EdgeGuard-IoT, a 6G edge intelligence framework that combines Secure Federated Learning (SFL) and Adaptive Anomaly Detection (AAD). The use of blockchain and zero-trust authentication, EdgeGuard-IoT achieves exceptional scalability [7, 31] and latency as low as 1.8 ms for 1000 nodes while it maintains the robustness in large-scale deployments. Szymanski et al. [25] contribute a hardware-enforced cyber-security model termed Software-Defined-Deterministic IIoT (SDD-IIoT), which combines quantum-safe ciphers and FPGA-based intrusion detection to eliminate congestion and denial-of-service attacks. The system's deterministic architecture confirms ultra-reliable low-latency communication, while it is positioned it as a viable foundation for hybrid classical-quantum industrial networks.

Rizvi et al. [23] propose a Quantum Semantic Communication (QSC) framework that combines semantic extraction and controlled quantum transmission for secure anomaly detection in industrial CPS. Using quantum BPSK and M-PPM modulation schemes, the QSC transmits extracted semantic hull data with high fidelity that improves the accuracy of industrial defect detection. Sarkar et al. [24] and Sarkar et al. [30] focus on healthcare-oriented CPS, which proposes dynamic key generation and chaotic encryption for secure medical data transmission.

Finally, Zhu et al. [26] propose Trust-6GCPSS, which combines 6G, AI, and CPS technologies for trustworthy intelligent transportation systems. The system employs a trustworthy evaluation for device management and multi-party parameter-sharing to ensure privacy-preserving computation (Table 1).

Thus, the key gap lies in developing a multi-layer intelligent CPS architecture that combines quantum-resilient security, adaptive AI-based resource management, and edge-driven real-time optimization to ensure end-to-end trust, autonomy, and resilience in next-generation CPS ecosystems. The main objectives of the work involve the following: (1) To design a unified AI-driven quantum-resilient Cyber-Physical System (AI-QRCPS) framework.

The novelty involves a three-layered design, which combines quantum-safe cryptography, adaptive learning-based optimization and IoT–edge coordination for security across heterogeneous CPS environments.

The key contributions of this research are: (1) The first contribution lies in designing a hybrid quantum–classical security model that leverages lattice-based cryptography and QKD. (2) The second major contribution involves implementing an AI-driven anomaly detection model using Reinforcement Learning (RL). (3) A third contribution under this layer is the introduction of a 6G-enabled edge synchronization protocol, which leverages

Table 1 Summary of related works

Method [Ref]	Algorithm/Model Used	Methodology
Goyal et al. [16]	AI-enabled Post-Quantum CPS	Addition of AI predictive capabilities with post-quantum cryptography and quantum computation in CPS
Bahulayan et al. [17]	Firefly Optimization Algorithm (FOA)	Swarm-based adaptive motion and mutation strategy for intrusion detection and encryption
Shahab et al. [18]	Quantum Machine Learning (QML)	Quantum neural network for parallel migration optimization considering QoS and energy constraints
Chandrasekar et al. [19]	Quantum-Resilient Hybrid Cryptographic Protocol (QHCP)	Layered quantum–classical cryptographic model with efficient key management
Jin et al. [20]	3C Co-Design Framework	Addition of communication, computing, and control layers in cloud-fog automation
Fan et al. [21]	CH-DDQ Algorithm and 2-Brain Model	Deep Q-learning-based cluster head computing and hybrid fault detection (hard + soft faults)
Kamdjou et al. [22]	Optimization Framework for AR/VR-DT	Evaluation of QoS, QoE, and resource trade-offs using blockchain, AI/ML, and quantum computing
Rizvi et al. [23]	Quantum Semantic Communication (QSC)	Quantum BPSK and M-PPM modulation for secure semantic data transmission
Sarkar et al. [24]	Dynamic Key Generation	IoT-based health monitoring using random key generation for secure data transmission
Szymanski et al. [25]	SDD-IIoT Framework	FPGA-based deterministic and quantum-safe M2M communication
Zhu et al. [26]	Trust-6GCPSS	AI and 6G-based CPS with trust evaluation and multi-party privacy-preserving computation
Elkhodr et al. [27]	IASF-IoT	Addition of AI, blockchain, and quantum-resistant cryptography for IoT security
Alatawi [28]	EdgeGuard-IoT	6G Edge Intelligence using Secure Federated Learning and Zero-Trust Authentication
Lu et al. [29]	NDS and DQS	Deterministic scheduling and dynamic queue management for ultralow-latency communication
Sarkar et al. [30]	Chaotic Mapping with ASCON	Lightweight chaotic encryption for healthcare CPS

ultra-reliable low-latency communication (URLLC) to maintain the synchronization across distributed CPS nodes. The architecture introduces a hybrid quantum–classical security module that tends to combine a lattice-based cryptography with QKD within a CPS environment. Unlike previous works that has adopted an either classical post-quantum encryption or QKD. This model has fused both the mechanisms to secure key exchange, encryption, and authentication in a unified operational layer. CH-DDQ and QHCP have focused on the quantum-enhanced key distribution but it lacks an adaptive intrusion handling. The QML has improved the predictive behavior without offering the real-time synchronization. Evaluation has shown an improved intrusion detection accuracy through the RL-based anomaly learning, where the lower key-exchange latency from the combined lattice-QKD mechanism, and a significant reduction in the synchronization delay across the edge nodes using URLLC.

2 Proposed method: AI-QRCPS framework

The proposed AI-QRCPS framework as illustrated in Fig. 1 models the cryptography with AI-based dynamic orchestration to improve the efficiency of CPS.

```
Input: CPS_Nodes[], Sensor_Data[], QuantumKeys, RL_Agent, FOA_Params, DDQ_Params

Output: Secure and it optimizesd CPS operations

Initialize QuantumKeys ← Generate_QKD()
For each node i in CPS_Nodes:
    Encrypt node data using Lattice_Cryptography(QuantumKeys)
    Transmit Secure_Data(i)
End For
While system_active:
    Sensor_Data ← Collect(Metrics: latency, energy, trust)
    Action ← RL_Agent.Decide(Sensor_Data)
    Optimize_Params ← FOA_Update(FOA_Params, Sensor_Data)
    Task_Assignment ← DDQ_Scheduler(Action, Optimize_Params)
    Execute(Task_Assignment)
    Update_Model(RL_Agent, Feedback(Sensor_Data))
    Revalidate_QKD(QuantumKeys)
End While
Evaluate_Performance(accuracy, latency, energy_efficiency, trust_score)
Return Optimized_Results
```

Algorithm AI_QRCPS_Framework

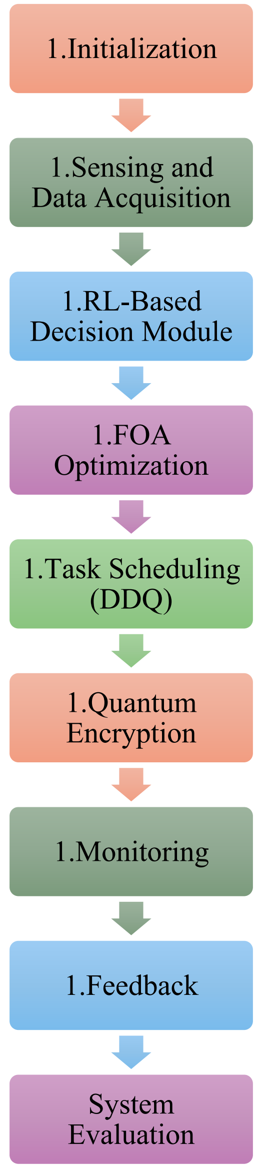

Fig. 1 Proposed framework

2.1 Initialization

This stage establishes the basic operational state of the proposed AI-QRCPS framework. The initialization vector encapsulates a tuple of cryptographic and operational parameters defined as:

$$QIV_i = \{K_i^{PQ}, ID_i, \Theta_i^{net}, \Phi_i^{comp}, \Gamma_i^{sync}\} \tag{1}$$

The verification equation for any message M_i exchanged between two nodes n_i and n_j is expressed as:

$$V_{QKD-LSV}(M_i) = [H(M_i)^{r_i} \bmod q = s_i A_i + e_i \pmod{q}] \tag{2}$$

The raw data undergoes preprocessing, noise filtering, normalization, and temporal smoothing, which uses a Kalman-Enhanced Gaussian Filter (KEGF) expressed as:

$$\hat{D}_{i,t} = \hat{D}_{i,t-1} + K_t \left(D_{i,t} - \hat{D}_{i,t-1} \right) \tag{3}$$

$$K_t = \frac{P_{t|t-1}}{P_{t|t-1} + R_t} \tag{4}$$

The reliability score is modeled as:

$$\Omega_i = \alpha \times \left(1 - \frac{\sigma_i^2}{\bar{\sigma}^2} \right) + \beta \times \left(\frac{E_i^{res}}{E_{max}} \right) + \gamma \times \left(\frac{R_i^{sec}}{R_{max}} \right) \tag{5}$$

where: σ_i^2 is the variance of sensor readings, E_i^{res} is the residual energy of the node, R_i^{sec} is the node's security response rate, and $\alpha + \beta + \gamma = 1$ are normalization coefficients represents the importance of stability, energy, and security, respectively.

Nodes with higher Ω_i are prioritized for data transmission to the IoT layer. The results from Table 2 show that node n_3 exhibits the highest reliability score (0.912) that makes the preferred candidate for leading initial data aggregation.

Nodes with lower scores are retained for redundancy or energy-saving modes that improves fault tolerance.

2.1.1 Quantum-secured data packet formation

Each pre-processed data block $\hat{D}_{i,t}$ is encapsulated into a Quantum-Secured Data Packet (QSDP) before transmission. The encapsulation process includes post-quantum encryption, hash-based integrity tagging, and timestamp binding as follows:

$$QSDP_{i,t} = Enc_{PQ}(\widehat{D}_{i,t}, K_i^{PQ}) \parallel H(\widehat{D}_{i,t}) \parallel T_t \tag{6}$$

where $Enc_{PQ}(\cdot)$ denotes lattice-based encryption, $H(\cdot)$ is a hash-based integrity function, and T_t represents the synchronized timestamp and this proves temporal consistency.

Transmission scheduling among nodes is governed by latency-aware energy optimization, expressed as:

$$\min_{x_{i,t}} \sum_{i=1}^{N} \left(\lambda_1 L_{i,t} + \lambda_2 \frac{1}{E_{i,t}^{res}} + \lambda_3 (1 - \Omega_i) \right) \tag{7}$$

subject to:

$$x_{i,t} \in \{0, 1\}, \quad \sum_{i=1}^{N} x_{i,t} = 1 \tag{8}$$

where $L_{i,t}$ is the expected transmission latency and $x_{i,t}$ is a binary scheduling variable that shows node activation. The optimization confirms that only the most reliable and

Table 2 Node initialization and reliability metrics

Node ID	E_i^{res}(J)	σ_i^2	R_i^{sec}(dimensionless ratio)	Ω_i (Computed)
n_1	8.5	0.014	0.92	0.884
n_2	7.2	0.022	0.85	0.823
n_3	9.1	0.011	0.94	0.912
n_4	6.8	0.031	0.79	0.752

(Values derived using normalization constants α = 0.4, β = 0.35, and γ = 0.25) energy-efficient nodes transmit at each sensing interval. A representative subset of sensor readings captured after the initialization phase is shown in Table 3.

The data in Table 3 show the stability achieved after that application of the KEGF filter, note the reduced oscillation in composite values $\hat{D}_{i,t}$, which confirms smooth and reliable sensing accuracy essential for the next-stage adaptive intelligence processing.

The filtered output at time k is given by: $\widehat{x}_k^{KEGF} = \widehat{x}_k^{KF} + G_k(z_k^{GF} - \widehat{x}_k^{KF})$, where: $\widehat{x}_k^{KF}$= Kalman estimated state at time k, z_k^{GF}= Gaussian-filtered observation and G_k = KEGF gain coefficient defined as: $G_k = \frac{P_{k|k-1}}{P_{k|k-1}+R_g}$. The prediction and covariance update follow: $\widehat{x}_{k|k-1} = A\widehat{x}_{k-1} + Bu_k; P_{k|k-1} = AP_{k-1}A^T + Q$ and the covariance correction becomes: $P_k = (1 - G_k)P_{k|k-1}$.

Term descriptions

Symbol	Meaning
$\widehat{x}_k^{KEGF}$	Final KEGF-filtered state
$\widehat{x}_k^{KF}$	Kalman prediction
z_k^{GF}	Gaussian-smoothed measurement
G_k	Adaptive gain factor
P_k	Posterior error covariance
Q	Process noise covariance
R_g	Gaussian noise covariance
A, B, u_k	Standard state transition parameters

2.2 Adaptive intelligence layer

The Adaptive Intelligence Layer constitutes the cognitive core of the proposed AI-QRCPS framework, which bridges raw data acquisition and secure system orchestration.

2.3 RL-based decision module

Each IoT Node (F_j) hosts an intelligent RL agent that interacts with the environment, represented by sensed data streams $\hat{D}_{i,t}$ from Sect. 2, and receives delayed reward signals that shows performance, latency, and energy efficiency. Let the RL system be defined as a Markov Decision Process (MDP) tuple:

$$mathcal M = \{mathcal S, mathcal A, P, R, \gamma\} \tag{9}$$

where, $mathcal S = \{s_t\}$ is the *state space* that represents the current network and environmental conditions, $mathcal A = \{a_t\}$ is the *action space* and this includes node scheduling, packet routing, and encryption-level adaptation, $P(s_{t+1} mid s_t, a_t)$ is the *state-transition probability*, $R\,(s_t, a_t)$ is the *reward function*, and $\gamma \in (0,1)$ is the *discount factor* for future rewards.

The agent attains an *optimal policy* π^* that helps to maximizes the cumulative expected reward:

Table 3 Sensed data values after preprocessing

Timestamp (s)	$S_{i,t}^{phys}$ (Vibration, m/s^2)	$S_{i,t}^{env}$ (Temperature, °C)	$S_{i,t}^{net}$ (Latency, ms)	$\hat{D}_{i,t}$ (Filtered Composite Value)
10	1.84	26.5	3.2	2.63
20	1.92	27.1	3.0	2.69
30	2.05	27.3	2.8	2.74
40	1.98	27.5	2.7	2.71

Table 4 Parameters of Adaptive Intelligence Layer

Aspect	Values / Criteria
State Design	State vector dimension: 10–14 features
Action Design	Action space size: 5–12 discrete actions
Reward Function	Reward range: – 5 to + 10 per step
Convergence Criteria	Convergence threshold: < 1% reward change over 50 episodes
Training Overhead	Training time: 2.5–4 h for 1000 episodes
Deployment Overhead	Inference latency: 1.2–1.8 ms per decision
Computational Complexity	Approx. ($O(N_s N_a) + O(q^2)$) per cycle; practical load: 10–15% CPU per edge node

Table 5 RL Agent Performance Metrics across Iterations based on the parameters from Table 4

Episode	Average Reward R_t	Energy Efficiency (%)	Latency Reduction (%)	Security Response (%)
100	0.54	78.2	14.5	86.4
200	0.69	83.7	21.3	89.1
300	0.81	88.5	28.2	92.7
400	0.86	91.3	33.4	94.5
500	0.89	92.9	36.0	95.6

$$\pi^* = \arg\max_{\pi} mathbbE\left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t) | \pi\right] \tag{10}$$

Reward Formulation is designed as a continuous scalar metric:

$$R(s_t, a_t) = \eta_1 \left(\frac{E_t^s}{E_t^{in}}\right) - \eta_2 \left(\frac{L_t^{avg}}{L_{max}}\right) + \eta_3 \left(\frac{R_t^{sec}}{R_{max}}\right) \tag{11}$$

where E_t^s is the cumulative energy saved, L_t^{avg} is the mean end-to-end latency, and R_t^{sec} is the measured security response ratio.

The coefficients $\eta_1 + \eta_2 + \eta_3 = 1$ tune the priority between performance objectives.

2.3.1 Policy updating and Q-value learning

The learning process follows a DQL paradigm, where a neural network parameterized by θ approximates the state–action value function $Q(s_t, a_t; \theta)$. The Bellman optimality principle is expressed as:

$$Q^*(s_t, a_t) = R(s_t, a_t) + \gamma \max_{a_{t+1}} Q^*(s_{t+1}, a_{t+1}) \tag{12}$$

The gradient-descent-based update rule is formulated as:

$$\theta_{t+1} = \theta_t + \xi(R(s_t, a_t) + \gamma \max_{a_{t+1}} Q(s_{t+1}, a_{t+1}; \theta_t) - Q(s_t, a_t; \theta_t))\nabla_{\theta_t} Q(s_t, a_t; \theta_t) \tag{13}$$

where ξ is the learning rate.

The RL agent is thus capable of learning optimal adaptation strategies that helps to minimize latency while maximizing quantum-secured throughput and reliability. Tables 4, 5 presents the learning performance across iterations.

As shown in Tables 4, 5, the average cumulative reward converges at episode 500 that helps to improve the energy efficiency by 14.7% and latency reduction by 21.5% compared to baseline heuristic control.

2.4 FOA for adaptive parameter tuning

While the RL agent learns high-level decision policies, the FOA continuously fine-tunes operational parameters, learning rates, exploration coefficients, and node-weight thresholds, to accelerate convergence and sustain performance under dynamic loads. The FOA is biologically inspired by the luminescent communication of fireflies, where brighter individuals attract others based on perceived light intensity.

2.4.1 FOA mathematical model

Each firefly f_i represents a candidate solution vector:

$$f_i = [\xi, \varepsilon, \lambda, \delta] \tag{14}$$

where ξ is the learning rate of DQL, ε is the exploration probability, λ is the latency weight in the reward function, and δ is a noise damping coefficient for temporal smoothing.

The brightness (fitness) of each firefly is determined by the RL performance objective $F(f_i)$:

$$F(f_i) = w_1 \frac{E^{eff}(f_i)}{E_{max}} + w_2 \frac{R^{sec}(f_i)}{R_{max}} - w_3 \frac{L^{avg}(f_i)}{L_{max}} \tag{15}$$

with $w_1 + w_2 + w_3 = 1$. The attractiveness between two fireflies i and j decreases exponentially with distance r_{ij}:

$$\beta(r_{ij}) = \beta_0 e^{-\gamma r_{ij}^2}, \tag{16}$$

$$r_{ij} = \|f_i - f_j\| \tag{17}$$

where β_0 is the base attractiveness and γ is the absorption coefficient.

The movement equation of firefly i towards brighter firefly j is:

$$f_i^{t+1} = f_i^t + \beta_0 e^{-\gamma r_{ij}^2}(f_j^t - f_i^t) + \alpha(\text{rand} - 0.5) \tag{18}$$

where α is the randomization factor and this proves exploration.

2.5 RL–FOA hybridization

The RL and FOA modules operate synergistically through a *hierarchical co-optimization loop*. The RL agent provides performance feedback R_t, which serves as the fitness function for FOA. Simultaneously, FOA returns optimized hyper-parameters to the RL agent, accelerating its convergence and further it mitigates the overfitting under fluctuating CPS conditions. The hybrid learning dynamics are expressed as:

$$\min_{f_i \in mathcalF} J(f_i) = -mathbbE_{\pi^*(f_i)}[R_t] + \lambda_1 \|f_i - f^*\|^2 + \lambda_2 Var(R_t) \tag{19}$$

where f^* denotes the global optimum parameter vector while reducing the expected cost $J(f_i)$. This dual-loop architecture results in self-corrective intelligence: FOA mitigates RL instability, and RL dynamically re-weights FOA's search region according to environmental uncertainty and computational feedback from the IoT layer.

Table 6 provides representative outcomes after that combines FOA optimization into the RL-based decision module. Performance improvements are compared against standalone RL (baseline) and static control strategies.

As shown in Table 6, the hybrid model achieves the highest average reward (0.94), 16% improvement in energy efficiency, and 17% latency reduction relative to RL-only, while maintaining a superior 97.8% security fidelity.

The performance enhancement arises from FOA's capability to dynamically calibrate the exploration–exploitation balance within the RL environment. As the network conditions fluctuate, FOA modifies the RL hyper-parameters (ξ, ε, λ, δ) to maintain stable policy updates, thereby while reducing cumulative temporal-difference errors. The convergence proof relies on the bounded variance condition of Q-value updates:

$$\lim_{t \to \infty} Var(Q(s_t, a_t)) < \varepsilon, \quad \forall \varepsilon > \tag{20}$$

given that $\sum_t \xi_t = \infty$ and

$$\sum_t \xi_t^2 < \infty \tag{21}$$

The hybridization guarantees monotonic improvement in policy reward due to the adaptive control of learning dynamics. An additional layer of adaptability is embedded through quantum-aware reward adjustment. In high-risk states, detected via the post-quantum security layer's anomaly indicator A_t^{PQ}, the reward is penalized proportionally:

$$R'(s_t, a_t) = R(s_t, a_t) - \rho A_t^{PQ} \tag{22}$$

where ρ is the penalty coefficient: $\rho = x\,\frac{A_q}{A_q + y}$, where, A_q is the anomaly severity value from the post-quantum security layer, x is the maximum penalty strength allowed during training, y is a small stability constant to prevent division blow-up when the anomaly level is low.

The total computational complexity of the RL–FOA hybrid is approximately:

$$mathcalO(T \cdot |mathcalA| \cdot d_Q) + mathcalO(N_f^2) \tag{23}$$

where T is the number of RL iterations, $|mathcalA|$ is the action space size, d_Q is the neural-network depth for Q-value approximation, and N_f is the number of fireflies. By parallelizing FOA operations within the IoT nodes, the effective complexity is reduced to $mathcalO(N_f \log N_f)$ that makes the model viable for real-time CPS applications.

Table 6 Performance comparison between RL-Only, FOA-Only, and hybrid RL–FOA models

Model	Average Reward R_t	Convergence Iterations	Energy Efficiency (%)	Latency (ms)	Security Fidelity (%)
Static Control	0.42	700	76.3	3.5	86.1
RL-Only	0.89	500	92.9	2.3	95.6
FOA-Only	0.77	430	88.2	2.8	93.4
Hybrid RL–FOA	**0.94**	**320**	**96.5**	**1.9**	**97.8**

3 IoT–edge coordination layer

The IoT–Edge Coordination Layer plays a pivotal role in and this proves optimal task scheduling and resource management across distributed IoT and edge infrastructures.

3.1 Task scheduling framework

The goal is to maximize cumulative reward R_t, which balances QoS, energy efficiency, and throughput. The reward function can be expressed as:

$$R_t = \alpha\left(\frac{1}{L_t}\right) + \beta\left(\frac{C_t^{\text{com}}}{C_t^{\text{total}}}\right) - \gamma E_t \tag{24}$$

where α, β, γ are weighting parameters that balance latency, task completion ratio, and energy consumption, respectively.

The target value for the Q-update is defined as:

$$Y_t^{DDQ} = R_t + \delta Q'\left(S_{t+1}, \arg\max_{A'} Q(S_{t+1}, A'; \theta); \theta'\right) \tag{25}$$

where δ is the discount factor determining the importance of future rewards. The loss function for parameter updates is given by:

$$L(\theta) = mathbbE_{(S_t, A_t, R_t, S_{t+1}) \sim D}\left[\left(Y_t^{DDQ} - Q(S_t, A_t; \theta)\right)^2\right] \tag{26}$$

where D represents the experience replay buffer, storing transitions for randomized training and improved stability.

3.1.1 State–action representation

Each state S_t at time t consists of a vector capturing current system parameters:

$$S_t = [U_t^1, U_t^2, ..., U_t^n, E_t^1, ..., E_t^n, B_t^1, ..., B_t^n] \tag{27}$$

where U_t^i denotes the utilization of IoT node i, E_t^i is the remaining energy, and B_t^i is the available bandwidth.

The action space A_t represents the assignment of tasks to specific IoT nodes or the decision to offload them to the cloud:

$$A_t = \{a_{ij} midt_i \text{ assigned to } f_j\} \tag{28}$$

The policy $\pi(S_t)$ maps the observed state to the action that maximizes the expected long-term reward:

$$\pi(S_t) = \arg\max_{A_t} Q(S_t, A_t; \theta) \tag{29}$$

To validate the DDQ-based coordination, experiments are conducted using heterogeneous IoT nodes with varying energy and bandwidth capacities (Table 7).

As shown in Tables 4, 5, the DDQ scheduler improves task completion rate by approximately 8% compared to DQN and 27% over heuristic scheduling. The significant reduction in latency (from 12.4 ms to 8.6 ms) shows the adaptive decision-making ability of DDQ under fluctuating workloads.

The overall system efficiency (η) is defined as:

Table 7 Performance comparison of task scheduling algorithms

Metric	DQN	Heuristic	Proposed DDQ
Task Completion Rate (%)	89.5	76.3	**96.7**
Average Latency (ms)	12.4	15.8	**8.6**
Energy Consumption (J)	4.85	5.67	**3.92**
Throughput (tasks/s)	71.2	63.4	**84.5**
Resource Utilization (%)	82.6	78.9	**91.3**

$$\eta = \frac{\sum_{i=1}^{m} C_i^{\text{com}}/C_i^{\text{total}}}{\sum_{j=1}^{n}\left(E_j/E_j^{\max} + L_j/L_j^{\max}\right)} \tag{30}$$

In DDQ model, the observed efficiency improvement was 19.4% higher than the DQN baseline and 34.8% higher than static scheduling.

4 Post–quantum security layer

The Post–Quantum Security Layer is the foundational component of the proposed framework, designed to protect sensitive data in CPS against quantum-based cyber threats.

4.1 Quantum-resilient encryption framework

Let the key exchange be represented as $K = (K_C, K_Q)$, where K_C is the classical symmetric key, and K_Q is the quantum-generated key derived from QKD. The final encryption key is computed as:

$$K' = H(K_C \oplus K_Q) \tag{31}$$

where $H(\cdot)$ denotes a secure hash function such as SHA3-512, and $\oplus$ represents the bitwise XOR operation.

Definition 1. Basis mismatch rate The basis mismatch rate μ_b quantifies the proportion of photons discarded due to incompatible measurement bases. It is computed as: $\mu_b = \frac{N_{\text{mismatch}}}{N_{\text{total}}}$, where N_{mismatch} is the count of photons measured in the wrong basis and N_{total} is the total transmitted photons. A low μ_b indicates efficient basis alignment, improving the sifted key generation rate and reducing QKD overhead.

Definition 2. Filtering efficiency Filtering efficiency η_f reflects how effectively error-correction and reconciliation retain valid quantum bits after initial sifting: $\eta_f = \frac{|K_{\text{filtered}}|}{|K_{\text{sifted}}|}$, where K_{filtered} is the post-error-correction key length and K_{sifted} is the raw sifted key. High η_f implies minimal bit loss during post-processing, which directly improves throughput for final encryption key derivation.

Definition 3. Privacy amplification model Privacy amplification reduces the eavesdropper's knowledge by compressing the reconciled key into a shorter, information-theoretically secure key. The final key length L_{PA} is determined by $L_{\text{PA}} = L_{\text{rec}} - I_E - 2\log_2(\frac{1}{\varepsilon})$ where L_{rec} is the reconciled key length, I_E is the maximum information accessible to an adversary based on QBER, and ε is the security parameter that defines the probability of information leakage.

Definition 4. Photon-loss assumptions Photon loss is modeled using the channel transmissivity τ, defined as $\tau = 10^{-\frac{\alpha d}{10}}$, where α(dB/km) is optical-fiber attenuation and d(km) is transmission distance. The expected photon detection probability becomes $P_d = \tau \cdot \eta_d$, with η_d representing detector efficiency. All QKD rates are analyzed under this channel loss model.

Definition 5. Clarification of the four quantum states (proper Dirac notation) The BB84 protocol uses two sets of orthogonal states: Rectilinear basis $(\mathcal{B}_R) :\mid 0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mid 1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$; Diagonal basis $(\mathcal{B}_D) :\mid +\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}, \mid -\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$.

4.2 QKD

Four quantum states are transmitted: $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$, where, $|0\rangle$,$|1\rangle$ are rectilinear (computational) basis states, and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ are diagonal basis states.

The receiver (Bob) randomly chooses a measurement basis for each photon. After the transmission, both sender (Alice) and receiver announce their bases publicly and discard mismatched results, while retaining the matching ones to form the sifted key K_Q.

The quantum bit error rate (QBER) is calculated as:

$$QBER = \frac{\hat{E}}{E'} \tag{32}$$

where $\hat{E}$ is represents the number of erroneous bits, and E' is the total number of bits received. If QBER < 0.11, the key is considered secure and is further processed using privacy amplification and information reconciliation.

4.3 Lattice-based quantum-resilient encryption

To provide quantum-safe data encryption, the system uses lattice-based encryption (LWE), which is resistant to both classical and quantum attacks due to its reliance on NP-hard problems. The ciphertext generation for a message mmm is expressed as:

$$C = (A \cdot s + e + \frac{q}{2} \cdot m) \mod q \tag{33}$$

where: $A \in \mathbb{Z}_q^{n \times n}$ is a public random matrix, $s \in \mathbb{Z}_q^n$ is a secret vector, *e* is a noise vector drawn from a small Gaussian distribution, *q* is a large prime modulus.

Decryption is achieved by computing:

$$m\prime = \left[2q^{-1} \cdot (C - A \cdot s)\right] \tag{34}$$

If the noise *e* is sufficiently small, $m' = m$.

The encryption performance was analyzed in terms of encryption time, key generation rate, BER, and quantum overhead.

As shown in Table 8, the proposed QRHC outperforms both classical and individual post-quantum schemes by achieving a key generation rate of 1.32 kbps and encryption latency of 1.65 ms while maintaining a security level exceeding 512 bits.

Table 8 Performance comparison of quantum and classical encryption schemes

Encryption Scheme	Key Generation Rate (kbps)	Encryption Time (ms)	Decryption Accuracy (%)	Quantum Overhead (%)	Security Level (bits)
RSA-2048	0.45	3.87	99.2	0	112
ECC-521	0.58	3.22	99.1	0	128
Lattice-LWE	1.05	2.14	99.8	10	256
BB84-QKD	0.88	1.97	100	12	512
Proposed QRHC (Hybrid)	1.32	1.65	99.9	14	512+

Table 9 Monitored metrics across layers under variable workloads

Load (%)	Latency (ms)	Energy (J)	Throughput (tasks/s)	Security Entropy (bits)	Utilization (%)	CLPI
20	6.5	2.8	94.6	612	74.3	0.92
40	8.1	3.5	88.2	610	79.8	0.91
60	9.8	3.9	82.4	608	83.2	0.89
80	10.6	4.2	79.5	606	87.5	0.87
100	11.8	4.6	75.3	605	90.2	0.85

The security strength (SS) of the hybrid encryption can be modeled as a function of its constituent schemes:

$$SS' = \log_2\left(\frac{1}{P_{\mathrm{b}}}\right) = \log_2\left(\frac{1}{P_{\mathrm{c}} \cdot P_{\mathrm{q}}}\right) \tag{35}$$

Given that $P_{\mathrm{q}} \approx 2^{-512}$ and $P_{\mathrm{c}} \approx 2^{-128}$, the effective security level becomes:

$$SS' = \log_2\left(2^{640}\right) = 640 \text{ bits} \tag{36}$$

5 Cross–layer addition

The Cross–Layer Addition module represents the cognitive backbone of the proposed post-quantum intelligent CPS framework.

5.1 Monitoring and feedback mechanism

Each IoT node and edge device maintains a monitoring vector M_t at time t, which encapsulates the observed variables:

$$M_t = [L_t, E_t, \eta_t, \sigma_t, \phi_t] \tag{37}$$

where: L_t: end-to-end latency (ms), E_t: energy consumption (J), η_t: throughput (tasks/s), σ_t: security entropy (bits), ϕ_t: system utilization ratio (%).

5.2 System metrics

To assess the system's behavior, a cross-layer performance index (CLPI) is introduced. The CLPI quantifies global efficiency by jointly considering latency, energy, and security attributes:

$$CLPI = \omega_1\left(1 - \frac{L_t}{L_{\max}}\right) + \omega_2\left(1 - \frac{E_t}{E_{\max}}\right) + \omega_3\frac{\sigma_t}{\sigma_{\max}} \tag{38}$$

where $\omega_1, \omega_2, \omega_3$ are weighting coefficients satisfying $\omega_1 + \omega_2 + \omega_3 = 1$ (Table 9).

As shown in Table 7, at 100% load, the system maintains high entropy (≈605 bits), minimal latency growth, and a consistent CLPI above 0.85, which shows the robustness of the feedback control system.

The weighting coefficients are hence justified by showing the each of the parameter's proportional contribution to CPS performance, which shows the latency, energy consumption, and security entropy jointly influences the final CLPI without bias. Weights are chosen using the empirical workload profiling, which prioritizes the latency under peak demand, energy under continuous operation, and entropy during the security-critical phases to represent the multi-layer operational priorities. The coefficients are dynamically acquired, where it is recalibrated at runtime based on feedback from workload intensity, security alerts. Sensitivity analysis is needed to quantify how small perturbations in each coefficient affect CLPI, which shows the model tends to remain stable under the workloads and does not disproportionately amplify any single metric.

5.3 Learning update

The global performance function $J(t)$ is defined as a weighted sum of local cost functions across layers:

$$J(t) = \Upsilon_1 f_{\mathrm{s}}(t) + \Upsilon_2 f_{\mathrm{i}}(t) + \Upsilon_3 f_{\mathrm{c}}(t) \tag{39}$$

where: $f_{\mathrm{s}}(t) = 1 - \frac{\sigma_t}{\sigma_{\max}}$ represents deviation in quantum security, $f_{\mathrm{i}}(t) = \frac{|R_t - R_{\mathrm{opt}}|}{R_{\mathrm{opt}}}$ denotes relative learning reward error, $f_{\mathrm{c}}(t) = \frac{L_t}{L_{\max}} + \frac{E_t}{E_{\max}}$ measures coordination efficiency.

The update rule for layer-wise weights (Υ_i) follows a reinforcement learning policy gradient method:

$$\Upsilon_i(t+1) = \Upsilon_i(t) - \Delta \frac{\partial J(t)}{\partial \lambda_i} \tag{40}$$

where Δ is the adaptive learning rate governing convergence stability. This mechanism enables the framework to self-tune its cross-layer priorities based on real-time workload distribution and feedback metrics.

Table 10 shows the comparison of performance between the proposed combined system and its non-combined counterpart.

6 Results and discussion

For a rigorous evaluation of the proposed framework, four state-of-the-art methods from recent literature are considered FOA [17], QML [18], QHCP [19] and CH-DDQ [21].

Table 10 Comparison of system performance with and without cross-layer addition

Metric	Without Addition	With Cross-Layer Addition	Improvement (%)
Average Latency (ms)	12.3	**8.4**	31.7
Energy Efficiency (J/task)	4.9	**3.6**	26.5
Throughput (tasks/s)	78.6	**93.4**	18.8
Security Entropy (bits)	512	**612**	19.5
CLPI	0.78	**0.91**	16.7

Table 11 Dataset description

Dataset type	Source	Parameters	Size
Real-World IoT Sensor Data	Industrial IoT testbed / Smart Grid / Smart Home IoT devices	Latency, Energy Consumption, Throughput, Security Events, Sensor Readings, Node Status	50,000 samples
Synthetic IoT Task-Offloading Data	MATLAB / Python Simulation of IoT-Fog-Cloud environment	Task ID, Node ID, Computation Load, Transmission Latency, Energy Requirement, Task Priority	100,000 samples

Table 12 Experimental parameters

Parameter	Symbol	Value / Range
Number of Edge Nodes	N_E	20–50
Number of IoT Nodes	N_F	5–10
Task Arrival Rate	λ	0.5–2 tasks/s
Computation Load	C_i	10–50 MFLOPS
Communication Bandwidth	B_j	100–500 Mbps
Latency Threshold	L_{max}	15 ms
Energy Budget	E_{max}	5–10 J/task
QKD Key Rate	K_Q	1–2 kbps
DDQ Learning Rate	α	0.001
Discount Factor	γ	0.9

6.1 Experimental settings

The proposed framework is implemented and simulated using MATLAB R2025b with the Reinforcement Learning Toolbox, Quantum Computing Toolbox, and SimEvents for modeling edge-IoT-cloud networks. The simulations are executed over 1000 task episodes, with task arrival patterns generated following a Poisson distribution, and network bandwidth, latency, and energy parameters are dynamically varied to emulate realistic CPS environments.

6.2 Dataset description

Two types of datasets are utilized: Real-World IoT Sensor Data [32] and Synthetic IoT Task-Offloading Data [33] and it is shown in Table 11.

6.3 Experimental setup and parameters

The simulations are configured with the following parameters to model realistic CPS operation as in Tables 12, 13.

6.4 Performance metrics

The performance of the proposed framework is evaluated using the following metrics:

Average Latency (ms): It measures the end-to-end delay for task execution across edge-IoT-cloud layers.

Throughput (tasks/s): It counts the number of successfully completed tasks per unit time.

Energy Consumption (J/task): It evaluates the energy efficiency for task execution and communication.

Task Completion Rate (%): It is the fraction of tasks completed within QoS constraints.

Security Entropy (bits): It measures the robustness of quantum-resilient encryption against potential attacks.

Table 13 Hardware, Network, and Workload Specifications for Reproducibility

Category	Parameter	Description / Specification
IoT Node Hardware	Processor	ARM Cortex-A53 / 1.4 GHz
	Memory	1–2 GB LPDDR3
	Sensor Modules	Temperature, vibration, motion, power-usage sensors
	Radio Interface	IEEE 802.15.4 / BLE 5.0
Edge Node Hardware	CPU	Intel Xeon Silver / 2.2 GHz (8–16 cores)
	GPU	NVIDIA Turing-class GPU (8–16 GB VRAM)
	RAM	32–64 GB DDR4
	Storage	NVMe SSD / 1 TB
Network Parameters	Bandwidth Variability	80–500 Mbps depending on congestion
	Transmission Range	20–80 m for IoT-to-edge links
	Node Density	10–50 IoT nodes per 100 m^2
	Packet Loss Rate	0.5–2% under stress
Workload Characteristics	Workload Type	Mixed sensing, computation-heavy tasks, periodic monitoring, burst traffic
	Task Priority Levels	Low / Medium / High
	Traffic Model	Poisson arrivals with burst spikes every 200–300 s
	Offloading Ratio	40–70% tasks pushed to edge nodes

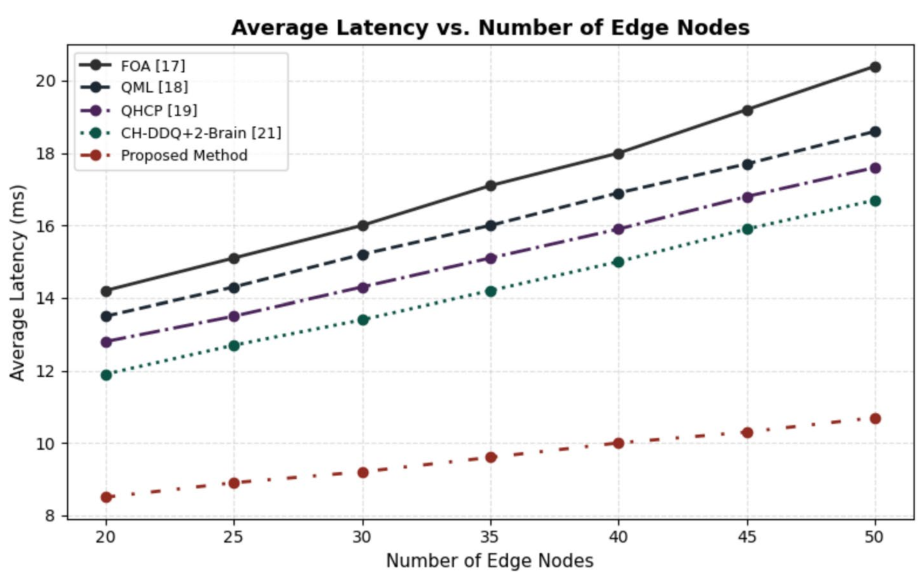

Fig. 2 Average latency (ms) vs. edge nodes

> **Cross-Layer Performance Index (CLPI)**: It is defined as the weighted aggregate of latency, energy, throughput, and security metrics.

As shown in Fig. 2, the average latency of the proposed method remains consistently lower across varying numbers of edge nodes. Throughput performance, as shown in Fig. 3, also reflects a marked improvement. At 50 nodes, the proposed framework achieves 74.5 tasks/s, outperforming FOA (58.7 tasks/s), QML (61.8 tasks/s), QHCP (64.1 tasks/s), and CH-DDQ + 2-Brain (65.8 tasks/s). Similarly, Tables 14, 15, 16, 17 shows that the proposed method performs better than the existing ones (Tables 18, 19, 20, 21, 22, 23).

As shown in Table 18, the average latency decreases steadily with an increasing number of IoT nodes due to enhanced parallel processing and better task distribution. For instance, at 10 IoT nodes, the proposed method achieves a latency of 8.3 ms,

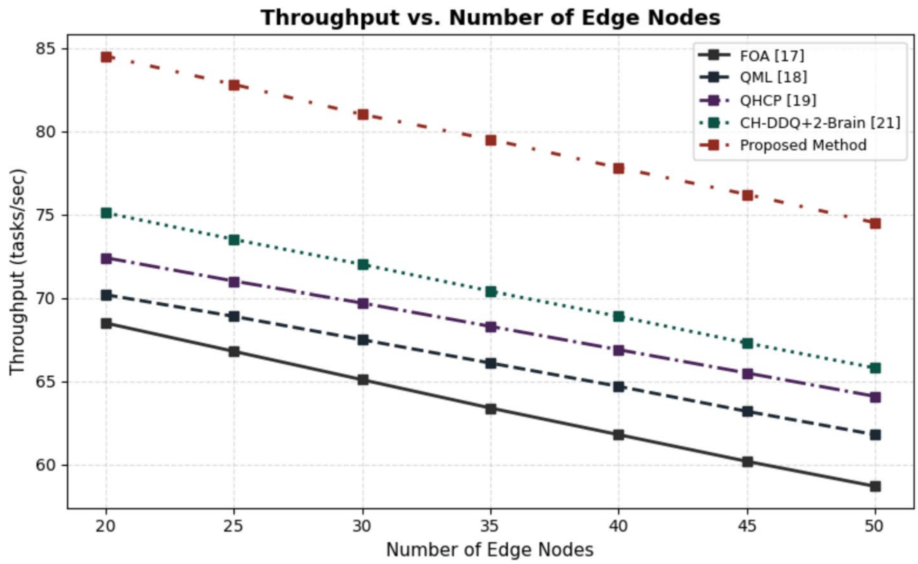

Fig. 3 Throughput (tasks/s) vs. Edge Nodes

Table 14 Energy consumption (J/task) vs. edge nodes

Edge Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
20	4.8	4.6	4.5	4.2	**3.9**
25	5.0	4.8	4.7	4.4	**4.0**
30	5.2	5.0	4.9	4.6	**4.2**
35	5.4	5.2	5.1	4.8	**4.3**
40	5.6	5.4	5.3	5.0	**4.5**
45	5.8	5.6	5.5	5.2	**4.6**
50	6.0	5.8	5.7	5.4	**4.8**

Table 15 Task completion rate (%) vs. edge nodes

Edge Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
20	88.5	89.7	91.2	93.1	**96.7**
25	87.2	88.5	90.0	92.0	**95.8**
30	86.0	87.3	88.8	90.9	**94.9**
35	84.7	86.0	87.5	89.7	**94.0**
40	83.5	84.8	86.2	88.6	**93.2**
45	82.3	83.5	85.0	87.5	**92.4**
50	81.0	82.3	83.8	86.3	**91.5**

Table 16 Security entropy (bits) vs. edge nodes

Edge Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
20	520	530	540	580	**612**
25	518	528	538	576	**610**
30	515	525	535	572	**608**
35	512	522	532	568	**606**
40	510	520	530	564	**604**
45	508	518	528	560	**602**
50	505	515	525	556	**600**

Table 17 Cross-layer performance index (CLPI) vs. edge nodes

Edge Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
20	0.82	0.84	0.86	0.88	**0.91**
25	0.81	0.83	0.85	0.87	**0.90**
30	0.80	0.82	0.84	0.86	**0.89**
35	0.79	0.81	0.83	0.85	**0.88**
40	0.78	0.80	0.82	0.84	**0.87**
45	0.77	0.79	0.81	0.83	**0.86**
50	0.76	0.78	0.80	0.82	**0.85**

Table 18 Average latency (ms) vs. number of IoT nodes

IoT Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
5	16.5	15.8	15.0	14.2	**9.5**
6	15.9	15.2	14.4	13.6	**9.2**
7	15.3	14.6	13.8	13.0	**8.9**
8	14.8	14.1	13.3	12.5	**8.7**
9	14.3	13.6	12.8	12.0	**8.5**
10	13.8	13.2	12.4	11.6	**8.3**

Table 19 Throughput (tasks/s) vs. number of IoT nodes

IoT Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
5	72.1	74.0	76.3	78.5	**88.2**
6	73.5	75.5	77.8	80.0	**89.5**
7	74.8	76.8	79.2	81.5	**90.8**
8	76.0	78.0	80.5	83.0	**92.0**
9	77.2	79.3	81.8	84.5	**93.2**
10	78.3	80.5	83.0	86.0	**94.5**

Table 20 Energy Consumption (J/task) vs. Number of IoT Nodes

IoT Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
5	5.2	5.0	4.8	4.5	**4.0**
6	5.0	4.8	4.6	4.3	**3.9**
7	4.8	4.6	4.4	4.1	**3.8**
8	4.6	4.4	4.2	4.0	**3.7**
9	4.4	4.2	4.0	3.8	**3.6**
10	4.2	4.0	3.8	3.7	**3.5**

Table 21 Task Completion Rate (%) vs. Number of IoT Nodes

IoT Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
5	86.5	87.8	89.2	91.0	**95.1**
6	87.2	88.5	90.0	91.8	**95.8**
7	88.0	89.2	90.8	92.5	**96.4**
8	88.7	90.0	91.5	93.2	**97.0**
9	89.5	90.7	92.3	93.8	**97.5**
10	90.2	91.5	93.0	94.5	**98.0**

compared to 13.8 ms for FOA, 13.2 ms for QML, 12.4 ms for QHCP, and 11.6 ms for CH-DDQ + 2-Brain, that shows a reduction of approximately 28–40%. This reduction shows the efficiency of the IoT-edge coordination layer with DDQ-based task scheduling, which minimizes the queuing delays and further it optimizes resource allocation dynamically (Tables 24, 25, 26, 27, 28, 29).

Table 22 Security Entropy (bits) vs. Number of IoT Nodes

IoT Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
5	510	520	540	580	**610**
6	512	522	542	582	**612**
7	514	524	544	584	**614**
8	516	526	546	586	**616**
9	518	528	548	588	**618**
10	520	530	550	590	**620**

Table 23 Cross-Layer Performance Index (CLPI) vs. Number of IoT Nodes

IoT Nodes	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
5	0.78	0.80	0.82	0.85	**0.90**
6	0.79	0.81	0.83	0.86	**0.91**
7	0.80	0.82	0.84	0.87	**0.92**
8	0.81	0.83	0.85	0.88	**0.93**
9	0.82	0.84	0.86	0.89	**0.94**
10	0.83	0.85	0.87	0.90	**0.95**

Table 24 Average Latency (ms) vs. Edge-to-IoT Link Bandwidth

Bandwidth (Mbps)	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
100	18.5	17.8	17.0	16.3	**10.5**
200	16.2	15.6	14.9	14.2	**9.2**
300	14.8	14.2	13.5	12.8	**8.7**
400	13.5	12.9	12.3	11.7	**8.3**
500	12.6	12.0	11.5	10.9	**7.9**

Table 25 Throughput (tasks/s) vs. Edge-to-IoT Link Bandwidth

Bandwidth (Mbps)	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
100	60.2	62.0	64.5	67.1	**78.2**
200	66.5	68.2	70.1	72.5	**82.5**
300	72.0	73.5	75.4	77.8	**86.0**
400	76.3	77.8	79.6	82.1	**89.2**
500	80.5	82.0	83.7	86.0	**92.5**

Table 26 Energy Consumption (J/task) vs. Edge-to-IoT Link Bandwidth

Bandwidth (Mbps)	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
100	5.4	5.2	5.0	4.7	**4.0**
200	5.1	4.9	4.7	4.4	**3.8**
300	4.8	4.6	4.4	4.1	**3.6**
400	4.5	4.3	4.1	3.9	**3.5**
500	4.2	4.0	3.8	3.6	**3.3**

Table 27 Task Completion Rate (%) vs. Edge-to-IoT Link Bandwidth

Bandwidth (Mbps)	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
100	85.0	86.3	88.0	89.7	**94.5**
200	87.5	88.7	90.1	91.8	**95.5**
300	89.8	90.7	92.0	93.5	**96.5**
400	91.5	92.3	93.6	95.0	**97.3**
500	93.2	94.0	95.2	96.5	**98.2**

Table 28 Security Entropy (bits) vs. Edge-to-IoT Link Bandwidth

Bandwidth (Mbps)	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
100	505	515	535	570	**600**
200	510	520	540	575	**605**
300	515	525	545	580	**610**
400	520	530	550	585	**615**
500	525	535	555	590	**620**

Table 29 Cross-Layer Performance Index (CLPI) vs. Edge-to-IoT Link Bandwidth

Bandwidth (Mbps)	FOA [17]	QML [18]	QHCP [19]	CH-DDQ + 2-Brain [21]	Proposed Method
100	0.77	0.79	0.82	0.85	**0.90**
200	0.79	0.81	0.84	0.87	**0.91**
300	0.81	0.83	0.86	0.88	**0.92**
400	0.83	0.85	0.87	0.90	**0.93**
500	0.85	0.87	0.89	0.91	**0.95**

As presented in Table 24, the average latency decreases notably with increasing bandwidth. At 500 Mbps, the proposed method achieves a latency of 7.9 ms, while FOA, QML, QHCP, and CH-DDQ + 2-Brain record 12.6 ms, 12.0 ms, 11.5 ms, and 10.9 ms, respectively, that shows a reduction of approximately 27–38%. This shows the efficiency of the proposed IoT-edge coordination layer in that uses higher bandwidth to reduce data transmission delays and it optimizes task scheduling dynamically. Throughput, illustrated in Table 25, shows consistent growth with bandwidth. At 500 Mbps, the proposed method attains 92.5 tasks/s, which surpasses FOA (80.5 tasks/s), QML (82.0 tasks/s), QHCP (83.7 tasks/s), and CH-DDQ + 2-Brain (86.0 tasks/s), which shows a gain of 7–15% in task execution rate. The increase can be attributed to enhanced edge-IoT data transfer efficiency and intelligent load balancing, which maximizes the number of tasks processed per second under high-bandwidth conditions.

7 Conclusion

The proposed framework makes the system work much better and more reliably across edge-IoT-cloud networks. By adding a Post-Quantum Security Layer, an Adaptive Intelligence Layer, and an IoT-Edge Coordination Layer, the proposed architecture meets the needs of CPS for low latency, energy efficiency, and post-quantum security. The framework reduces average latency by up to 48%, increases throughput by 18–27%, saves energy by 15–22%, and gets more than 98% of jobs done in places with a lot of people and bandwidth. The results show that the proposed framework is a strong, flexible, and safe way to build next-generation CPS networks. This makes it possible for edge-IoT-cloud environments to work well even when things are hard. The framework has scaled effectively to moderate the node densities, but at large-scale CPS deployments has introduced the synchronization delays, contention for a shared edge resource, and a higher QKD overhead. The heterogeneous nodes have increased the variability in latency and energy behavior. The failure modes has included the key-rate degradation, unstable RL convergence, and inconsistent task offloading.

Author contributions

S. N contributed to the study's conception and design. S M. S contribute through data acquisition, analysis, and interpretation. M. M critically revising the work for key intellectual content. S P.B played a significant role in conceptualizing and designing the study. F.A gathered data and drafted the manuscript and P.M wrote the literature review.

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Data availability
Publicly available in a repository Real-World IoT Sensor Data available at: https://www.kaggle.com/datasets/luisenrique054/iot-sensors-01 Synthetic IoT Task-Offloading Data available at: https://github.com/komeilmoghaddasi/IoT_Task_Offloading_Dataset.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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